

Gradient

(1)

→ If a scalar function $f(x, y, z)$ is defined and differentiable at all points in some region, then f is a differentiable scalar field.

$\vec{\nabla}$, del operator may be applied to scalar fields and $\vec{\nabla}f$ is a vector field.

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \quad (\text{del operator})$$

$$\vec{\nabla} f = \text{gradient of } f$$

$$= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

Gradient

(2)

In term of physics, the rate of change with respect to distance of variable quantity as temperature or pressure, in the direction of maximum change

For this we need to understand a scalar field. A scalar field is something that has a particular value at every point in space. For e.g temperature.

A vector field is same as a scalar field but except for only having value at every point in space, it has a value and direction at every point in space.

Problems (gradient) (3)

Q1. if $f = xy^2 - yz$ find gradient

$$\text{Sol. } \nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$= \frac{\partial}{\partial x} (xy^2 - yz) \hat{i} + \frac{\partial}{\partial y} (xy^2 - yz) \hat{j} + \frac{\partial}{\partial z} (xy^2 - yz) \hat{k}$$

$$\nabla f = y^2 \hat{i} + (2xy - z) \hat{j} - y \hat{k}$$

Ans

(4)

Q2. If $f(x, y) = x + y^2$ find $\text{grad } f$

$$\begin{aligned}\text{Sol. } \nabla f &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \\ &= \frac{\partial}{\partial x} (x + y^2) \hat{i} + \frac{\partial}{\partial y} (x + y^2) \hat{j}\end{aligned}$$

$$= (1 + 0) \hat{i} + (0 + 2y) \hat{j}$$

$$\nabla f = \hat{i} + 2y \hat{j}$$

Ans

Q3. Find $\text{grad } f$ if $f = \sqrt{x^2 + y^2}$

$$\text{Sol. } \nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

(5)

Q3. Solution:

$$\text{grad } f = \vec{\nabla} f = \frac{\partial}{\partial x} (x^2 + y^2)^{1/2} \vec{i} + \frac{\partial}{\partial y} (x^2 + y^2)^{1/2} \vec{j}$$

$$= \frac{1}{2} (x^2 + y^2)^{\frac{1}{2} - 1} \cdot \frac{\partial}{\partial x} (x^2) \vec{i} + \frac{1}{2} (x^2 + y^2)^{\frac{1}{2} - 1} \cdot \frac{\partial}{\partial y} (y^2) \vec{j}$$

$$= \frac{1}{2} (x^2 + y^2)^{-1/2} (2x) \vec{i} + \frac{1}{2} (x^2 + y^2)^{-1/2} (2y) \vec{j}$$

$$= \frac{x}{\sqrt{x^2 + y^2}} \vec{i} + \frac{y}{\sqrt{x^2 + y^2}} \vec{j}$$

Q4. Find grad f .

$$\text{if } f = 3x^2 \sqrt{y} + \cos(3z)$$

Sol.

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

$$\nabla f = 3y^{1/2} \frac{\partial}{\partial x} x^2 i + 3x^2 \frac{\partial}{\partial y} y^{1/2} j + \frac{\partial}{\partial z} (\cos 3z) k$$

$$\nabla f = 3y^{1/2} (2x) i + (3x^2) \frac{1}{2} y^{-1/2} j - 3 \sin 3z k$$

$$\nabla f = 6x\sqrt{y} i + \frac{3x^2}{2\sqrt{y}} j - 3 \sin 3z k$$

Ans

Curl and Divergence (7)

Roughly speaking divergence measures the tendency of a fluid to collect or disperse at a point, and curl measures the tendency of the fluid to swirl around the point.

Divergence is scalar that is a single number, while the curl is itself a vector.

Let $\vec{V}(x, y, z)$ be differentiable vector function

$$\vec{V}(x, y, z) = V_1 \vec{i} + V_2 \vec{j} + V_3 \vec{k}$$

$$\text{div } \vec{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

is called the divergence of $\vec{V}$ (8)

$$\vec{\nabla} = \text{del operator} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Curl is a vector that measures the rotation or circulation of a vector field.

The curl of vector field

$$\vec{F}(x, y, z) = f(x, y, z) \hat{i} + g(x, y, z) \hat{j} + h(x, y, z) \hat{k}$$

is denoted by $\text{curl } \vec{F}$ and is

given by

$$\text{curl } \vec{F} = \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

$$\text{curl } \vec{F} = \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right) \hat{i} + \left(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial x} \right) \hat{j} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \hat{k}$$

PROBLEMS SOLVING

9)

Q1. Calculate the divergence of the vector field $\vec{F} = x\hat{i} + y^3z^2\hat{j} + xz^3\hat{k}$

Solution:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial x}{\partial x} + \frac{\partial}{\partial y} (y^3z^2) + \frac{\partial (xz^3)}{\partial z}$$

$$\vec{\nabla} \cdot \vec{F} = 1 + 3y^2z^2 + 3z^2x$$

Ans

Q2. Find the divergence of vector field?

$$\vec{F}(x, y, z) = x^2y\hat{i} + 2y^3z\hat{j} + 3z\hat{k}$$

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} (x^2y) + \frac{\partial}{\partial y} (2y^3z) + \frac{\partial}{\partial z} (3z)$$

$$= 2xy + 6y^2z + 3 \text{ Ans}$$

(12)

Q3. Find the divergence of

$$F(x, y) = \frac{x}{y} \hat{i} + (2x - 3y) \hat{j}$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} \left(\frac{x}{y} \right) + \frac{\partial}{\partial y} (2x - 3y)$$

$$\nabla \cdot \vec{F} = \frac{1}{y} - 3 \quad \text{Ans}$$

Q4. Find divergence of

$$\vec{G} = x^2 \hat{i} + 2z \hat{j} - y \hat{k}$$

$$\nabla \cdot \vec{G} = \frac{\partial}{\partial x} x^2 + \frac{\partial}{\partial y} (2z) + \frac{\partial}{\partial z} (-y)$$

$$\nabla \cdot \vec{G} = 2x + 0 + 0$$

Ans

Q5. Find divergence of $\vec{F} = x\hat{i} + y\hat{j}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y = 1 + 1 = 2$$

Ans

Q6. Find divergence of $\vec{G} = \frac{4y}{x^2} \hat{i} + \sin y \hat{j} + 3\hat{k}$

Sol. $\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} \left(\frac{4y}{x^2} \right) + \frac{\partial}{\partial y} (\sin y) + \frac{\partial}{\partial z} (3)$

$$= 4y \left(-\frac{2}{x^3} \right) + \cos y$$

$$= -8y x^{-3} + \cos y$$

Ans

$$\left(\frac{\partial y}{\partial x} = \frac{\partial}{\partial x} \left(\frac{1}{x^2} \right) \right)$$

$$= -2x^{-3}$$

Q7. Find divergence of $\vec{G} = e^x \hat{i} + \ln xy \hat{j} + e^{xyz} \hat{k}$

$$\vec{\nabla} \cdot \vec{G} = \frac{\partial}{\partial x} e^x + \frac{\partial}{\partial y} (\ln xy) + \frac{\partial}{\partial z} e^{xyz}$$

$$= e^x + \frac{1}{y} + e^{xyz} (xy)$$

$$= e^x + \frac{1}{y} + xy e^{xyz}$$

Ans

Curl

As $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$, $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$ (12)

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) - \hat{j} \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$\vec{\nabla} \times \vec{F}$ is Sometimes called
rotation of $\vec{F}$

CROSS PRODUCT

$$\vec{a} \times \vec{b} = ?$$

$$\text{Let } \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$= \hat{i} (a_y b_z - a_z b_y) - \hat{j} (a_x b_z - a_z b_x) + \hat{k} (a_x b_y - a_y b_x)$$

(15)

PROBLEMS

Q₁. Find the curl of $\vec{F}(x, y, z) = 3x^2\vec{i} + 2z\vec{j} - x\vec{k}$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 & 2z & -x \end{vmatrix}$$

$$= \left\{ \frac{\partial(-x)}{\partial y} - \frac{\partial(2z)}{\partial z} \right\} \vec{i} - \left\{ \frac{\partial(-x)}{\partial x} - \frac{\partial(3x^2)}{\partial z} \right\} \vec{j} + \left\{ \frac{\partial(2z)}{\partial x} - \frac{\partial(3x^2)}{\partial y} \right\} \vec{k}$$

$$= (0-2)\vec{i} - (-1-0)\vec{j} + (0-0)\vec{k}$$

$$\vec{\nabla} \times \vec{F} = -2\vec{i} + \vec{j}$$

Ans

Q2. Find curl of $\vec{F} = y^3 \hat{i} + xy \hat{j} - z \hat{k}$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} \frac{\partial}{\partial x} & \hat{j} \frac{\partial}{\partial y} & \hat{k} \frac{\partial}{\partial z} \\ y^3 & xy & -z \end{vmatrix}$$

$$= \hat{i} \left\{ \frac{\partial(-z)}{\partial y} - \frac{\partial(xy)}{\partial z} \right\} - \hat{j} \left\{ \frac{\partial(-z)}{\partial x} - \frac{\partial(y^3)}{\partial z} \right\} \\ + \hat{k} \left\{ \frac{\partial(xy)}{\partial x} - \frac{\partial(y^3)}{\partial y} \right\}$$

$$\vec{\nabla} \times \vec{F} = 0 \hat{i} - 0 \hat{j} + (y - 3y^2) \hat{k}$$

$$\vec{\nabla} \times \vec{F} = (y - 3y^2) \hat{k} \quad \text{Ans}$$

Q3. Compute curl of $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix}$$

$$= (-2y - 2y)\vec{k} = -4y\vec{k}$$

Ans